Measure, Predict, Prove: Testable AI Decision Support for Urban Planning

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Title	Measure, Predict, Prove: Testable AI Decision Support for Urban Planning (72/80)
Abstract	A rezoning changes homes, emissions and access, may never be built, and is weighed differently by each stakeholder. Drawing on six years of PhD research in urban planning, AI and GIS, we present a decision support workflow that splits the planner's questions: what would a change do, is it likely, what would do better, and was it right. Every answer is checked in ArcGIS Pro against public records such as permits. In three Manhattan rezonings, full build-out capacity was 8,423 homes and 561 were built by 2025: capacity is not a forecast. Planners leave with tools and a checklist. 584/600 characters. Published as written in the agenda.
Keywords	planning support systems; ArcGIS Pro; redevelopment forecasting; model evaluation; AI agents
Topic category	State and local government: Planning (focus area: Efficiencies and Innovations in Planning)
Audience and level	Planners, GIS analysts in planning and development agencies, urban researchers. Intermediate.
Primary speaker	Parya Payami, Department of Architecture, University of Wisconsin-Milwaukee (six years of PhD research in urban planning, AI and GIS)

Expanded abstract

Problem. Planners weigh rezonings and permits with capacity estimates, forecasts and impact figures. Checking those numbers against what was later permitted and built is slow: the records sit in different systems, are located differently, and lag construction by years. Without that check, a convincing map can carry an error nobody sees.

The system: five planning questions, each with its own tool and its own test. The first four come from PlanArena, our parcel-level planning model (4.99 million parcels, ten US cities); the fifth runs in ArcGIS Pro. Each output says what evidence stands behind it, and untested outputs are labelled as such.

Planning question	Tool	Evidence (Manhattan)	Limit stated
1. Measure: what would a change do?	Impact engine, 11 indicators with sensitivity ranges	Held-out lots 2015-25: homes r = 0.79, emissions r = 0.60	Totals overstated 1.47x; no skill for jobs; access and runoff untested
2. Predict: is it likely?	Redevelopment model using only data available at the decision year	Out-of-time AUC 0.85-0.90 over 15 origins (paper)	Ranks lots better than it predicts how many
3. Recommend: what would do better?	Search within zoning and precedent	Plausibility filter: share of recommended lots that redeveloped rose from 11% to 21%	Best plan found, not a proven optimum
4. Decide: who would choose it, under which rule?	Nine voting rules over simulated interests	Rules never all agreed in 12 districts	Illustrative: assumed preferences, not public support
5. Prove: was it right?	ArcGIS Pro workflow scoring a prediction layer against later public records	Main case below	Records have their own gaps; each source lists its known traps

Where ArcGIS does the work. In ArcGIS Pro, Spatial Join and Generate Near Table match records to parcels, definition queries score subsets, and error maps show where a forecast missed; Network Analyst and Spatial Analyst serve the supporting cases. PlanArena scenarios export to ArcGIS Urban; the evaluation runs in Pro.

A version anyone can play. PlanArena runs as a browser Studio and a phone game: players paint uses onto real parcels, watch the indicators change, and a simulated council then votes between their plan, the recommended plan and no change. Play it at planarena-manhattan.onrender.com/play (free server: the first visit can take a minute to wake). In the talk, the audience tries one scenario on their phones before ArcGIS Pro shows what was actually built.

Proof that the workflow works. Main case: did Manhattan redevelop where a 2020 forecast said it would? Outcome records postdate the forecast and were never used to build it.

Test (2020 forecast, 42,170 lots)	Observed records, 2021-2025	Forecast	Soft-site rule	What it shows
Where construction started	DOB NOW new-building first permits (150 lots), matched by tax lot (BBL), then nearest lot	AUC 0.81; top 10% of lots had 5.6x the base rate	AUC 0.81	Ranking works, but a planners' rule of thumb ranks starts as well
Where buildings were completed	Certificates of occupancy for new buildings (215 lots), lots without a 2015+ building	AUC 0.67	AUC 0.39, below chance	The rule of thumb fails for completions; the forecast holds

What the checks changed. Certificates are geocoded to the street frontage: only 30 of 601 near Manhattan lots fell inside one, so a plain overlay counted 16 completed lots instead of 430. Certificates also lag starts by years: 53% of certified lots already had a recent building in 2020, so they were excluded. Condominium billing lots needed location matching. **What it does not show:** housing totals (the outcome definitions differ from the model's), or cause and effect. The capacity figure in the abstract is full build-out, an end state; likelihood times capacity gave 607 homes against 561 built, with wide model ranges.

Supporting evidence (completed; in the talk only briefly):

Supporting evidence	Result	Read as
Fire coverage, NYC	Posted-speed drive times were a median 2.2 min faster than FDNY travel times (167 areas); 92% within 1 min after a bias estimated on the other areas (leave-one-out)	A modeled coverage map needs an observed correction before it guides siting
Permit persistence, NYC and Chicago	Past permits ranked next period's solar (NYC) and new-construction (Chicago) areas: Spearman 0.97 and 0.94	A baseline any forecast should beat; counts were not evaluated
Pipeline checks	A field holding the true outcome scores 1.0; random fields score 0.08 (178 areas) and -0.22 (77 areas)	Software checks, not evidence of predictive skill
AI agent evaluation	32 tasks, a fresh agent per task, no access to reference answers: all met pre-registered checks. Of 10 trap requests, 3 refused, 5 redirected, 2 completed with a warning; the agents found 6 flaws in our analysis, since fixed	One run per task, graded by the author; repeated runs are next

Fifteen-minute plan.

Minutes	Content
0-2	The problem in one Manhattan district: what zoning allows, what was forecast, what got built
2-5	Measure, predict, recommend for that district in the PlanArena Studio, with each output's evidence and limit
5-6	The audience plays: paint a scenario on their phones and see its impacts and the simulated council's vote
6-11	Prove, live in ArcGIS Pro: score the forecast against permits, see the geocoding and lag traps change the result, map the misses
11-13	An AI agent runs the same check and declines a request that would leak the outcome
13-15	Limits and the one-page checklist
15-20	Questions

Limits. Impacts are validated in Manhattan only; agreement with past records is not causation; ranking skill is not accurate totals; votes rest on assumed preferences; usability with planners is not yet evaluated. Test data are public; with an AI agent, prompts and outputs go to the model provider.

Takeaways. A tested ArcGIS Pro workflow for checking a planning forecast against later permits and certificates, notes on 16 public outcome sources and their traps, and a one-page checklist. Code and results: github.com/ParyaPayami/proveit. Data: NYC MapPLUTO, DOB NOW filings and certificates of occupancy, FDNY dispatch, Chicago permits; PlanArena manuscript (Payami, 2026).